



Anterior Abdominal Wall — MCQ Practice

Anatomy GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. The costal margin palpated during abdominal examination is formed by the cartilages of which ribs?

EASY

- A. 1st-6th
- B. 7th-10th
- C. 11th-12th
- D. 5th-7th only
- E. 8th-12th

2. Which layer, present in most regions of the body, is ABSENT from the anterior abdominal wall?

EASY

- A. Superficial fascia
- B. Fascia transversalis
- C. Extraperitoneal fat
- D. Deep fascia
- E. Parietal peritoneum

3. Which dermatome is marked by the umbilicus?

EASY

- A. T10
- B. T7
- C. T12
- D. L1
- E. T4

4. At which vertebral level does the xiphoid process lie?

EASY

- A. T4
- B. T7
- C. T9
- D. L1
- E. T12

5. A radiologist reviews an axial CT at the lower border of L1, through the tip of the 9th costal cartilage. Which structure lies on this plane?

MEDIUM

- A. Sigmoid colon
- B. Umbilicus
- C. Bifurcation of the aorta
- D. Appendix
- E. Fundus of the gallbladder

6. During hernia surgery in an obese man, the surgeon finds that the outer layer of the superficial fascia is fatty and varies in thickness with nutrition. Which layer is this?

MEDIUM

- A. Camper's fascia
- B. Scarpa's fascia
- C. Fascia transversalis
- D. External oblique aponeurosis
- E. Rectus sheath

7. A layer of superficial fascia continues into the scrotum, loses its fat and forms a fascia containing smooth muscle. Which fascia is formed?

MEDIUM

- A. Camper's fascia
- B. Scarpa's fascia
- C. Dartos fascia
- D. Buck's fascia
- E. Colles' fascia

8. After a straddle injury, urine from a ruptured penile urethra spreads into the scrotum, penis and lower abdominal wall but not into the thigh. Why?

MEDIUM

- A. Camper's fascia is attached to the inguinal ligament
- B. Scarpa's fascia is attached to the fascia lata
- C. The deep fascia of the abdomen blocks it
- D. Fascia transversalis seals the inguinal canal
- E. The dartos muscle contracts around the urine

9. During hernia repair the surgeon sees the posterior rectus sheath end at a curved margin, leaving a weaker, hernia-prone area. Where is this area?

MEDIUM

- A. Around the linea alba
- B. Lateral to the linea semilunaris
- C. Medial to the inguinal ligament
- D. Below the arcuate line
- E. Above the transpyloric line

10. A 30-year-old woman has colicky umbilical pain with abdominal distension and altered bowel habit. Which structure is most likely involved?

MEDIUM

- A. Sigmoid colon
- B. Right kidney
- C. Gallbladder
- D. Appendix
- E. Small intestine

11. Which structure is formed by the thickened, in-rolled lower free border of the external oblique aponeurosis?

MEDIUM

- A. Inguinal ligament
- B. Conjoint tendon
- C. Linea semilunaris
- D. Arcuate line
- E. Cremaster muscle

12. The conjoint tendon inserting into the pubic crest and pecten pubis is formed by which muscles?

MEDIUM

- A. External oblique and internal oblique
- B. Rectus abdominis and pyramidalis
- C. Internal oblique and transversus abdominis
- D. External oblique and transversus abdominis
- E. Transversus abdominis and rectus abdominis

13. The cremaster muscle covering the spermatic cord is derived from which muscle?

MEDIUM

- A. External oblique
- B. Transversus abdominis
- C. Rectus abdominis
- D. Internal oblique
- E. Pyramidalis

14. A patient has dilated superficial veins above the umbilicus. Into which vein do they normally drain?

MEDIUM

- A. Femoral vein via the great saphenous vein
- B. Axillary vein via the lateral thoracic vein
- C. Portal vein directly
- D. Internal iliac vein
- E. Inferior epigastric vein only

15. A surgeon makes a paramedian incision midway between the costal margin and the umbilicus. Which layers form the anterior wall of the rectus sheath at this level?

HARD

- A. External oblique aponeurosis only
- B. Aponeuroses of all three flat muscles
- C. Transversus abdominis and fascia transversalis
- D. Costal cartilages 5–7
- E. External oblique aponeurosis fused with the anterior lamina of the internal oblique

16. An indirect hernia passes through an oval opening about 1.2 cm above the mid-inguinal point. In which layer is this opening, and what covering does that layer give the spermatic cord?

HARD

- A. Fascia transversalis; internal spermatic fascia
- B. External oblique; external spermatic fascia
- C. Internal oblique; cremasteric fascia
- D. Scarpa's fascia; dartos
- E. Transversus abdominis; conjoint tendon

17. A surgeon chooses a horizontal rather than a vertical incision on the lower abdomen to get a thinner scar. Which anatomical fact supports this?

HARD

- A. The deep fascia runs horizontally
- B. The rectus sheath is thicker horizontally
- C. The tendinous intersections are horizontal
- D. Langer's (cleavage) lines of the abdominal wall run horizontally
- E. Scarpa's fascia is absent below the umbilicus

18. At the level of the xiphoid, above the costal margin, what lies directly behind the rectus abdominis?

HARD

- A. Fascia transversalis only
- B. Posterior lamina of internal oblique
- C. The 5th, 6th and 7th costal cartilages
- D. Transversus abdominis aponeurosis
- E. External oblique aponeurosis

19. Which pair of horizontal planes divides the abdomen into the nine regions, and at what levels?

HARD

- A. Transpyloric (L1) and transumbilical (L3/4)
- B. Transumbilical (L3/4) and intertubercular (L5)
- C. Transpyloric (L1) and intercostal (L4)
- D. Subcostal (L1) and transtubercular (L3)
- E. Subcostal (L3) and intertubercular (L5)

20. A bodybuilder's 'six-pack' shows transverse grooves across rectus abdominis. Where are the three constant tendinous intersections?

HARD

- A.** Pubic symphysis, umbilicus, xiphoid
- B.** Tip of the xiphoid, umbilicus, and midway between them
- C.** Costal margin, arcuate line, pubic crest
- D.** Only below the umbilicus
- E.** At each of the 7th–10th costal cartilages

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	The costal margins are formed by the cartilages of the 7th, 8th, 9th and 10th ribs.
2	D	EASY	The anterior abdominal wall has no deep fascia, which lets it distend after meals and in pregnancy.
3	A	EASY	The umbilicus marks the T10 dermatome.
4	C	EASY	The xiphoid process, at the apex of the subcostal angle, lies at the level of T9.
5	E	MEDIUM	This is the transpyloric plane: pylorus, neck of pancreas, DJ flexure, gallbladder fundus, renal hila, origins of the SMA and portal vein, and the conus medullaris.
6	A	MEDIUM	Camper's fascia is the superficial fatty layer of the superficial fascia below the umbilicus; Scarpa's is the deeper membranous layer.
7	C	MEDIUM	In men Camper's fascia loses its fat in the scrotum and becomes the dartos fascia with smooth muscle.
8	B	MEDIUM	Urine collects deep to Scarpa's fascia; its attachment to the fascia lata closes the space off from the thigh.
9	D	MEDIUM	The arcuate line is the lower free edge of the posterior sheath. Below it only fascia transversalis lies behind rectus, so the wall is weaker.
10	E	MEDIUM	The umbilical region contains the jejunum and ileum, mesentery and transverse colon; colicky pain with distension points to the small intestine.
11	A	MEDIUM	The inguinal ligament (ASIS to pubic tubercle) is the rolled lower border of the external oblique aponeurosis.
12	C	MEDIUM	The arching fibres of the internal oblique fuse with those of transversus abdominis to form the conjoint tendon.
13	D	MEDIUM	The cremaster muscle and fascia come from the lower arched fibres of the internal oblique.
14	B	MEDIUM	The umbilicus is a venous watershed: above it blood drains via the lateral thoracic vein to the axillary vein; below it via the superficial epigastric and great saphenous veins to the femoral vein.
15	E	HARD	Between the costal margin and the arcuate line the internal oblique aponeurosis splits: its anterior lamina joins external oblique in front; its posterior lamina joins transversus behind.
16	A	HARD	The deep inguinal ring is in the fascia transversalis, which forms the internal spermatic fascia around the cord.
17	D	HARD	Langer's lines on the anterior abdominal wall run horizontally, so incisions along them heal with a thinner scar.
18	C	HARD	Above the costal margin the anterior wall is external oblique aponeurosis only and the posterior wall is the 5th-7th costal cartilages.
19	E	HARD	The subcostal plane (10th costal cartilage, L3) and the intertubercular plane (iliac tubercles, L5), with the two midclavicular lines, make the nine regions.
20	B	HARD	The three tendinous intersections lie at the xiphoid tip, at the umbilicus and midway between; a fourth may occur below the umbilicus.